EV Charging Infrastructure Optimization

Group Number: D07

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Introduction

- The rapid global transition to Electric Vehicles (EVs) requires a massive, coordinated scale-up in charging infrastructure.
- Building charging stations is expensive and resource-intensive. Current deployment is often reactive rather than strategic, leading to dense clusters in some areas and "charging deserts" in others.
- Without careful planning, infrastructure expansion can worsen geographic and demographic inequalities.
- **Significance:** By leveraging Operations Research (OR) methodologies, we aim to transform subjective infrastructure planning into a rigorous, data-driven mathematical process to ensure efficiency and fairness.

Problem Statement

- **Core Question:** Given a limited budget and an existing network of EV chargers, how can we expand the infrastructure to maximize both geographic coverage and regional fairness?

Sub-Problem 1: The Coverage Problem

Where should new high-capacity charging hubs be placed to serve the highest number of existing charging stations?

Sub-Problem 2: The Fairness Problem

How should a fixed budget of new charging connectors be allocated across administrative regions to ensure underserved areas receive adequate infrastructure?

- Simple heuristic approaches fail to consider these strict budget constraints and fairness trade-offs simultaneously. An OR framework is essential.

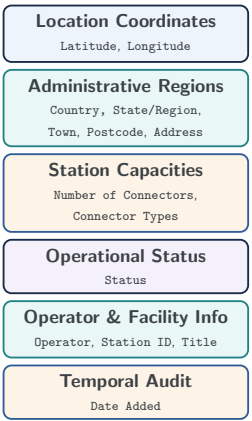
Objectives

- To mathematically model the EV charging network expansion as a set of linear and integer programming problems.
- To identify optimal locations for new hubs to maximize geographic coverage under a strict budgetary constraint.
- To formulate a fairness-driven allocation strategy that maximizes the minimum connectors-per-station ratio across regions.
- To systematically analyze and preprocess a massive, real-world observational dataset to derive parameters for these models.
- To build an interactive, data-driven decision support dashboard that allows stakeholders to run these mathematical models in real-time.

Dataset: Global EV Charging Stations

Dataset Overview

- Public infrastructure dataset sourced from **Kaggle** (*Rishi Panchal*).
- **Dataset Size:** Comprises **10,000 records** spanning **14 parameters/features**.
- We are focusing on a dataset of **10,000 charging-station records** comprising **14,565 connectors**.
- **Why this dataset?** It provides a highly realistic snapshot of infrastructure, necessary for spatial and capacity-based optimization.



Dataset: From Raw Data to OR Parameters

The raw records will be transformed into the parameters required by the two optimization models.

Geospatial Data

`lat` (Latitude)
`lon` (Longitude)

Used for candidate hub clustering and distance metrics

$\Downarrow$

d_{ij}

Haversine distances

Capacity Data

`num_connectors`
`status`

Physical capacity of each operational station

$\Downarrow$

C_i

Current regional capacity

Regional Data

`Country, State/Region`

Administrative boundaries for resource targeting

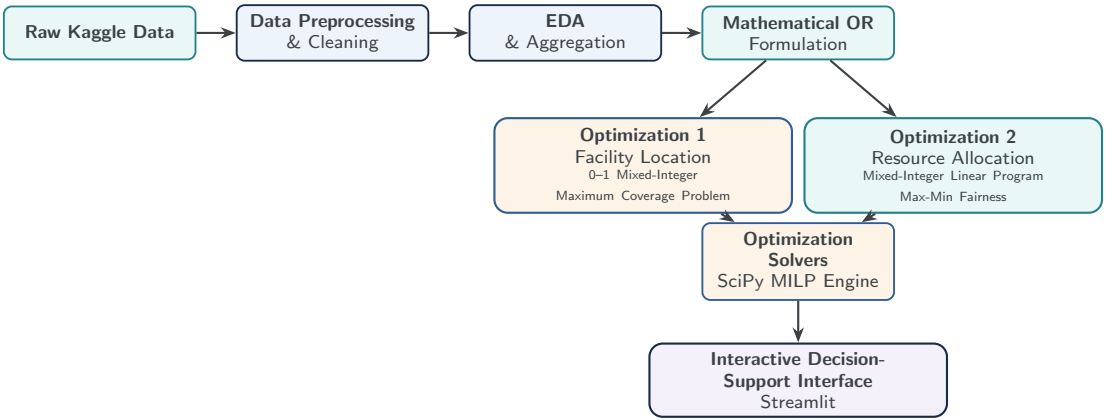
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Groupings

Regional aggregation

Raw Data $\rightarrow$ Exploratory Data Analysis (EDA) $\rightarrow$ OR Models

Proposed Methodology



Optimization 1: Hub Location

Mixed-Integer Programming: Facility Location (Maximum Coverage)

Raw Data → OR Parameters Translation

Input Data: Latitude and longitude of coverage stations after preprocessing $\xrightarrow{\text{Spherical K-Means}}$ M Candidate Hubs (j).
Distance Matrix: $d_{ij} = \text{Haversine}(\text{station}_i, \text{hub}_j) \implies N_i = \{j \mid d_{ij} \leq R\}$ (Hubs within radius R).

Decision Variables

$y_j \in \{0, 1\}$: Select candidate hub j ($j = 1 \ldots M$).

$z_i \in \{0, 1\}$: Binary coverage state of station i ($i = 1 \ldots N$).

Objective Function

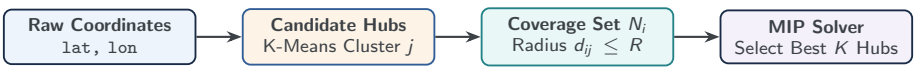
$$\max Z = \sum_{i=1}^{N} z_i$$

Maximize total existing stations covered by at least one selected hub.

Constraints

1. Hub Limit Constraint:
$\sum_{j=1}^{M} y_j \leq K$ (Max K hubs)

2. Coverage Constraint:
$z_i \leq \sum_{j \in N_i} y_j \quad \forall i$



Optimization 2: Connector Allocation

Mixed-Integer Linear Programming: Resource Allocation (Max-Min Fairness)

Raw Data → OR Parameters Translation

Grouping: `Country + State/Region` → Administrative regions ($i = 1 \dots R$).
Operational Stations (S_i): Number of operational stations in region i. **Operational Connectors ($C_{i,\text{old}}$):** Existing operational connectors in region i.

Decision Variables & Objective

$x_i \in \mathbb{Z}_{\geq 0}$: New connectors allocated to region i.

$t \in \mathbb{R}$: Continuous floor variable (min network CPS).

Objective Function:

$$\boxed{\max Z = t}$$

Maximize the lowest Connectors-Per-Station (CPS) ratio across all regions to enforce equity.

Constraints

1. Total Connector Budget:

$$\sum_{i=1}^{R} x_i \leq B \quad \text{(Total new connectors} \leq B\text{)}$$

2. Max-Min Fairness Floor Constraint:

$$\frac{C_{i,\text{old}} + x_i}{S_i} \geq t \quad \forall i \in \{1 \dots R\}$$

Post-allocation CPS of every region must be $\geq t$.

Raw Capacity ($C_{i,\text{old}}, S_i$) → Max-Min Allocation (x_i) → Network Equity (t)

Tools & Technologies

Python
Core Development

Pandas + NumPy
Data Processing

SciPy
OR Modeling

Scikit-Learn
Clustering

Git / GitHub
Version Control

Streamlit
Interactive UI

Plotly
Geospatial Mapping

Data Processing → OR Modeling →
Solver Optimization → Web Dashboard

Conclusion & Next Steps

- **Summary:** The proposed framework applies rigorous Operations Research to strategic EV charging expansion, prioritizing both coverage efficiency and geographic fairness.
- **Current Status:** The Kaggle dataset has been identified, and the proposed MIP/MILP formulations have been designed for the two optimization objectives.
- **Next Steps:**
  - Perform data preprocessing and EDA
  - Extract parameters for both optimization models
  - Implement and validate the optimization models using SciPy
  - Perform sensitivity analysis for budget and coverage radius
  - Develop the Streamlit decision-support interface
  - Generate deployment recommendations